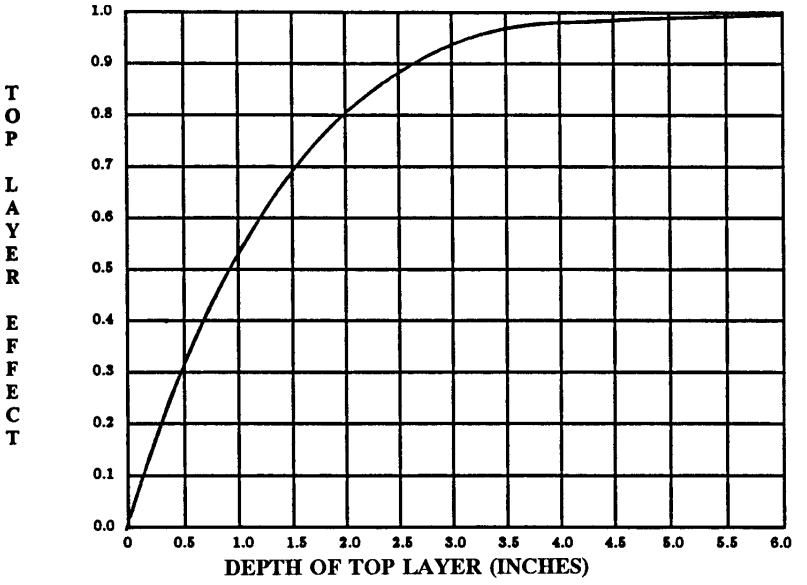


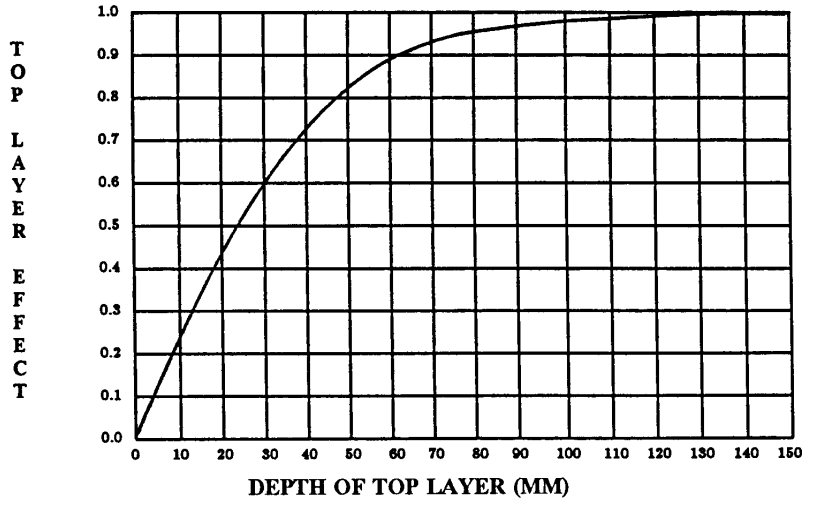




Figure 4. Backscatter Surface Density Effects

*(Top Layer Effect Curves)*

# Moisture

The Model 3430 gauge uses a 40 mCi (1.48 GBq) Americium-241:Beryllium neutron source to measure the hydrogen content (consequently the water content) of the material.

Neutrons emitted by the Am-241:Be source penetrate the material and are *thermalized* (or slowed). *Thermalization* is the process where neutrons are slowed to the point where further collisions with hydrogen or other materials will not continue to slow the neutron.

The Model 3430 gauge contains a helium-3 neutron detector that is sensitive to thermalized neutrons. This detector is insensitive to non-thermalized or "fast" neutrons and, as a result, the counts obtained are directly proportional to the amount of hydrogen/moisture present in the material.

The depth of measurement, or depth at which 98% of the counted neutrons pass before reaching the detector, is a function of moisture content:

$$\textbf{\textit{Depth (in.)} = 11 – (0.17 × M), where: M = moisture in pcf}$$

or

$$\textbf{\textit{Depth (mm)} = 280 - (0.27 × M), where: M = moisture in kg/m}^3$$

Therefore, the higher the moisture content in the material being measured, the smaller the depth of measurement. The normalized curve set shown in Figure 5 on the next page illustrates the effect of moisture content on the depth of measurement.

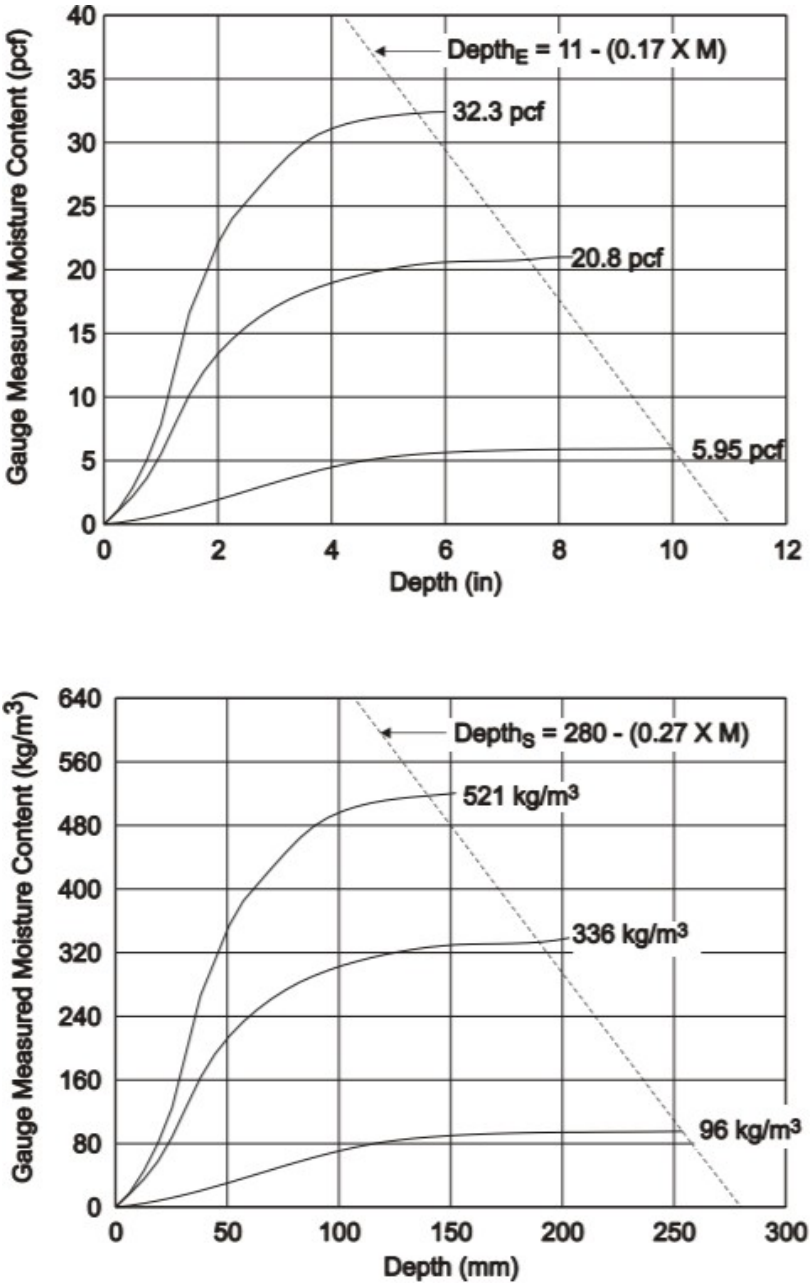



Figure 5. Effect of Moisture on Depth of Measurement

# Chapter 3:
# Using the Gauge

This chapter covers the following topics and tasks:

- ✓ Basic operation of the Model 3430
- ✓ Initial setup
- ✓ Preparing the test site
- ✓ Taking measurements

# The Keypad

The keypad of the Model 3430 Surface Moisture-Density Gauge consists of eleven keys—an eight-function keypad plus the ⟨**YES**⟩, ⟨**NO/ESC**⟩, and ⟨**LIGHT**⟩ keys.

The gauge is equipped with an internal beeper to verify keystrokes. If a *beep* is not heard when a key is pressed, the keystroke was not recognized and should be repeated.

The ⟨**YES**⟩ and ⟨**NO/ESC**⟩ keys are used for responses to specific questions displayed on the screen. The ⟨**LIGHT**⟩ key is used to turn the LCD backlight on and off.

The up and down arrows allow the operator to scroll through various function lists displayed by the gauge.

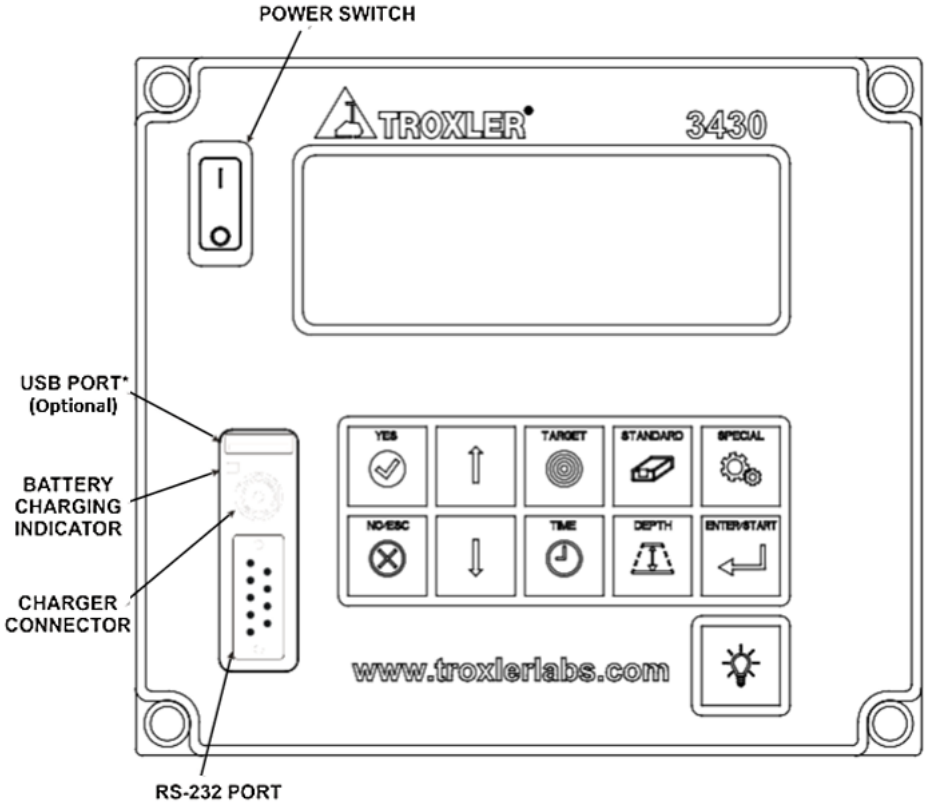



Figure 6. Model 3430 Keypad

Table 1 provides a detailed description of the individual keys.

Table 1. Model 3430 Keypad Functions

| KEYS | DESCRIPTION |
|---|---|
| **YES** | Answers *Yes* to prompts. |
| **NO**<br>**ESC** | Answers *No* to prompts and exits menus without saving changes. |
| ↑ | Scrolls the display up. |
| ↓ | Scrolls the display down. |
| **TARGET** | Allows entering or enabling of a Proctor or Gmb value. |
| **TIME** | Allows the operator to change the count time. |
| **STANDARD** | Use to access the *Standard Count* mode. |
| **DEPTH** | Allows entry of the source rod depth. |
| **SPECIAL** | Provides access to the *Special* functions. |
| **START**<br>**ENTER** | Starts a measurement or completes answer entry. |
| **LIGHT** | Turns on and off the LCD display backlight. |

## NOTE

**The gauge will automatically turn off after five hours if no keys are pressed.**

# Source Rod Positions

As shown in Figure 7, the source rod can be placed in the **SAFE**, backscatter, or direct transmission positions. When not taking measurements, keep the source rod in the **SAFE** position. When measuring thin layer or other materials through which you cannot drill a hole, use the backscatter position. In the direct transmission positions, the source rod extends into a pre-drilled hole.

Figure 7 shows the available positions of the source rod.

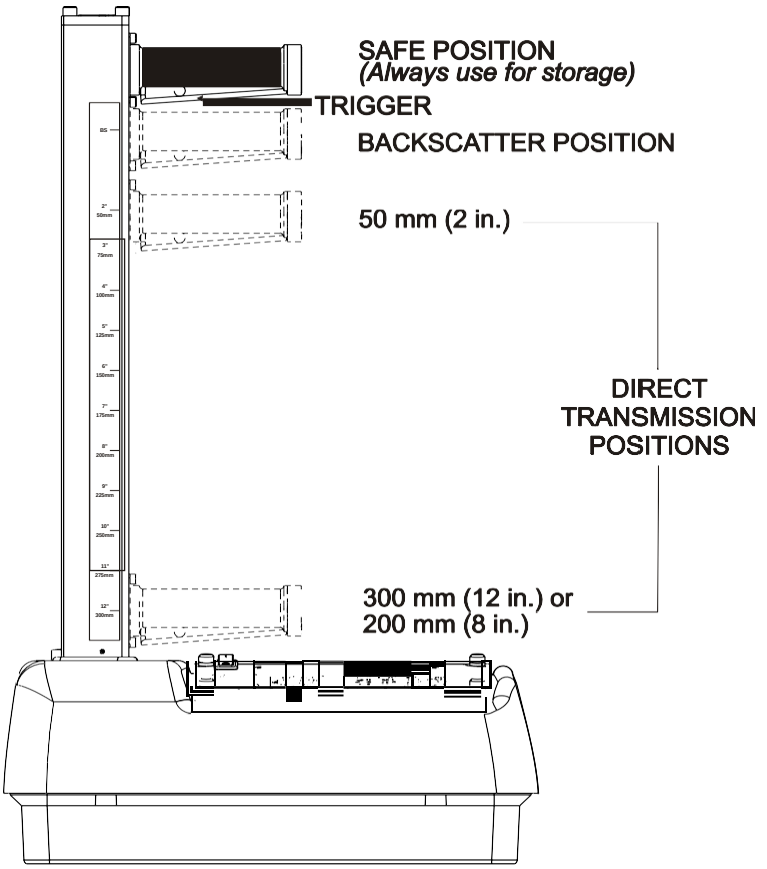



Figure 7. Source Rod Positions

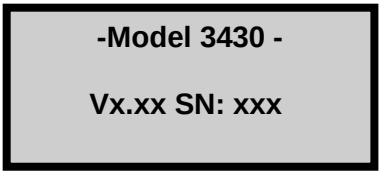
# Daily Inspection

The gauge should be inspected daily before use to ensure proper operation of all safety features. Refer to Appendix D for the inspection procedure.

# Turning the Gauge On

The gauge uses rechargeable NiMH batteries (included) as a power source. When first turned on, the control panel displays test characters before proceeding to the self-test.

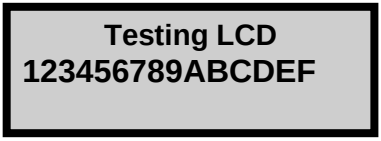
To turn the gauge on, toggle the on/off switch located to the left of the gauge's display. Upon turning the gauge on, the gauge displays:



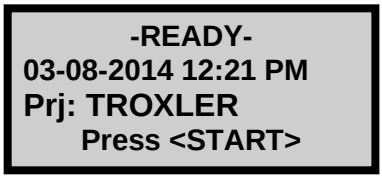

**NOTE**

**When turning on the Gauge a prompt will appear to choose a set language. Select desired language using the up and down arrows and press ENTER/START. If the incorrect language is set refer to chapter 5.**

The gauge then performs a test of its LCD (liquid crystal display):




After the gauge performs a 300-second self-test, the gauge enters the *Ready* mode. In this state any of the gauge functions may be accessed. The *Ready* mode display is:



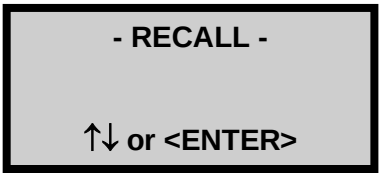

The first line of the display indicates the current *count time*. The second line of the display indicates the selected source rod depth.

# Gauge Setup

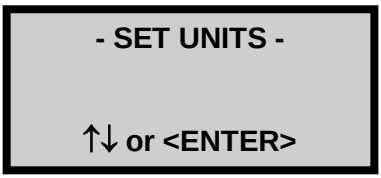
After unpacking the gauge and turning it on, several parameters can be initialized, such as measurement units and count time. These parameters do not usually change once they are set.

## Setting Measurement Units

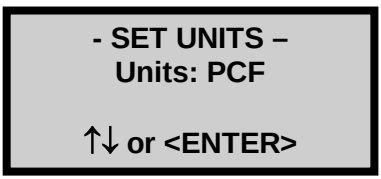
The Model 3430 gauge allows measurement results to be displayed in either metric or US units. To set the measurement units, first access the *Special* function menu by pressing ⟨**SPECIAL**⟩. The gauge displays:




Press the down arrow seven times to display:




To select *Set Units* press ⟨**START/ENTER**⟩.



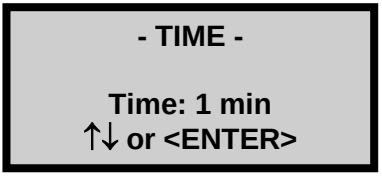

Use the up and down arrows to scroll through the available units. When the desired units are displayed press ⟨**START/ENTER**⟩.

## Setting the Count Time

The *count time* defines how long the gauge measures. Longer count times produce better measurement precision. Troxler

recommends a count time of one minute for most sample measurements.

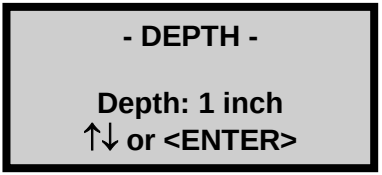
To change the count time, press ⟨**TIME**⟩ to display:




Use the up and down arrows to scroll through the available count times. When the desired count time is displayed press ⟨**START/ENTER**⟩.

## Setting the Depth

To change the depth of measurement, press ⟨**DEPTH**⟩.




Use the up and down arrows to scroll through the available measurement depths. When the desired depth is displayed press ⟨**START/ENTER**⟩ to select the displayed depth and return to the *Ready* mode.

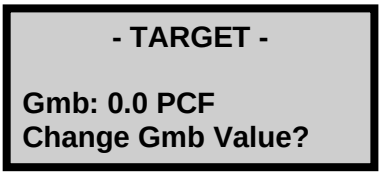
## Selecting the Mode (Gmb/Proctor)

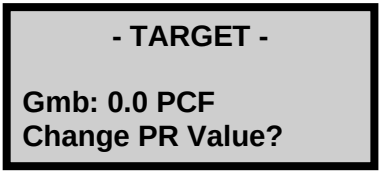
The gauge may be used on construction materials (soils, asphalt, concrete, and so on). To select the *Soil* mode, enter or activate a Proctor value. To select the *Asphalt* mode, enter or activate a Gmb value. Only one Gmb and one Proctor can be stored in the gauge at one time.

**NOTE**

**To measure concrete, use either the Asphalt or Soil mode. For moisture results select the Soil mode. For density measurement only, use the Asphalt mode.**

To enter or activate a Gmb or Proctor value, press ⟨**TARGET**⟩. The display will be one of the following:




To switch from a Gmb value to a Proctor value, or vice versa, use the arrow keys.

- To activate the displayed value, press ⟨**NO/ESC**⟩.
- To change the displayed value, press ⟨**YES**⟩. The first digit of the value will flash. Use the arrow keys to scroll through the possible entries (0 – 9) and when the correct value for the current digit is displayed, press ⟨**START/ENTER**⟩. The gauge will proceed to the next digit to the right.

When the value entry is complete, the gauge activates the value and returns to the *Ready* mode.

# Taking the Standard Count

The Model 3430 gauge uses a Cesium-137 and an Americium-241:Beryllium source for taking measurements. These radioactive sources undergo a natural decay process, resulting in a gradual loss in the intensity of their radiation. The time required for the source strength to diminish by 50% is referred to as the *half-life*.

To compensate for the source decay and to check proper operation of the gauge, a daily reference *standard count* should be performed. To ensure the highest accuracy possible with the gauge, it is important to take a daily standard count.

The gauge is equipped with a reference standard block for taking the standard count. Place the reference standard block on a dry, flat surface at least 3 m (10 ft.) from any large vertical surface and at least 10 m (33 ft.) from any other radioactive source. The surface should be asphalt, concrete or soil at least 10 cm (4 in.) thick and with a density of at least 1600 kg/m$^3$ (100 pcf). The right side of the gauge, farthest from the handle, should be against the metal butt plate (see Figure 8).

To begin the standard count procedure, press ⟨**STANDARD**⟩.

```
- STANDARD –

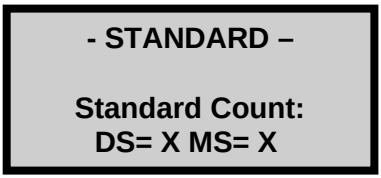
DS= 1870 MS= 468
New Std Cnt?
```

To take a new standard count, press ⟨**YES**⟩**.**

```
- STANDARD –

Press START for
Standard Count
```

Ensure that the gauge is positioned as shown in Figure 8. To initiate the standard count press ⟨**START/ENTER**⟩. After the count is complete, the display is:




Troxler recommends that the operator keep a daily log of the moisture and density standard counts (see Appendix E for a sample log you can use). To verify gauge stability, compare the daily standard count to a reliable reference as follows:

- During the first four days of operation of a new or recalibrated gauge, compare the daily standard count to the factory calibrated values.
- After the first four days of operation (or after taking four standard counts), compare the daily standard count to the average of the last four counts. Acceptable standard count limits are:

±1% each day for DS (density standard) and

±2% each day for MS (moisture standard).

After recording the standard counts, press ⟨**YES**⟩ to return to the *Ready* mode.

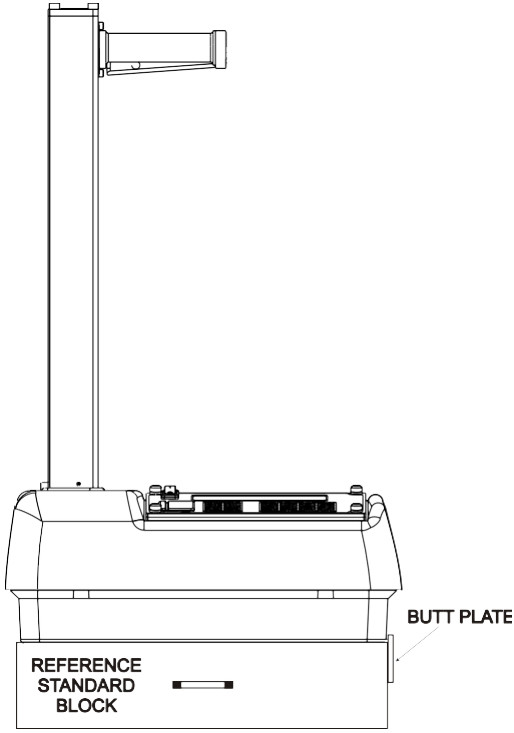



Figure 8. Standard Count Position

# Site Preparation

Preparation of the test site surface is critical to gauge performance. This section provides site preparation procedures for both soils and base courses and asphalt surfaces. To ensure the most accurate gauge readings, the appropriate preparation procedure should be followed.

## Soil and Base Course Preparation

1. Locate a level site free from any large holes, cracks, or debris (soil surface conditions are critical to accurate measurements).
2. Smooth the surface by moving the scraper plate in a back and forth motion. Filler such as fine sand may be used to fill in the surface voids.

**NOTE**

**Use only enough filler to fill the voids. Too much filler will cause an error in the measurement.**

3. For direct transmission measurements, put the drill rod through the extraction tool and then through one of the guides on the plate (see Figure 9).
4. **Wearing a radiation badge and safety glasses (or other locally approved safety devices),** step on the plate and hammer the drill rod at least 50 mm (2 in.) deeper than the desired test depth. The drill rod increments include the additional depth.
5. Remove the drill rod by pulling straight up and twisting the extraction tool. **Do not loosen the drill rod by tapping from side to side with a hammer.** This will distort the hole or cause loose material to fall into the hole.
6. To ensure accurate placement of the gauge, before removing the scraper plate mark the test area using the drill rod as shown in Figure 10.

7. Carefully pick up the scraper plate and place the gauge on the surface prepared by the plate. Insert the source rod into the hole made by the drill rod. **Use care when inserting the source rod; do not to disturb the soil around the hole.**
8. Lower the source rod into the hole. Release the trigger and lock the source rod into the correct position. A *click* should be heard when the source rod is locked into position.
9. Gently slide the gauge toward the keypad so the source rod makes contact with the wall of the hole.

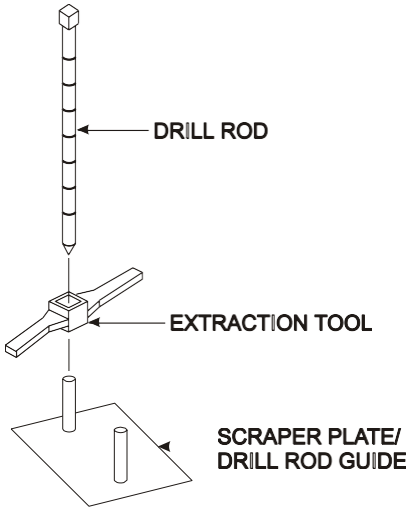



Figure 9. Drill Rod Positioning

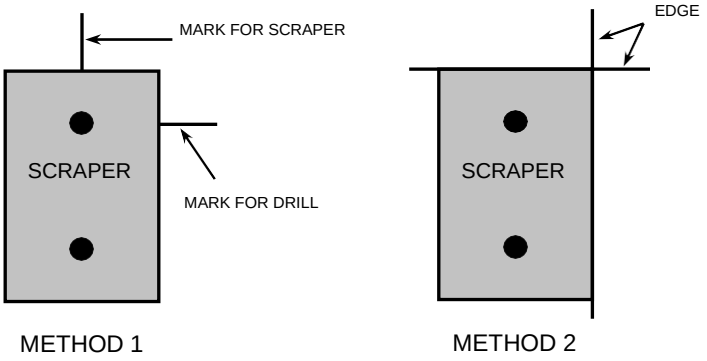



Figure 10. Marking the Test Area

## Asphalt Surface Preparation

It is possible, but usually not necessary, to take direct transmission readings on asphalt. Drilling a hole in asphalt can be difficult and may require the use of a drill (rather than the drill rod) if the asphalt has cooled and hardened.

Under normal conditions, a backscatter reading provides an accurate measurement of asphalt density.

1. Find a smooth location on the asphalt. The operator may want to fill the voids on open mixes with sand or cement. Take care to leave the asphalt exposed. **The gauge base must rest on the asphalt, not the fill material!**
2. Ensure that the gauge does not "rock." It must remain steady. If rocking occurs, find a more suitable test site. If taking a measurement around a core, the gauge may be moved up to a few inches to any side of the hole.

# Taking a Measurement – Soil Mode

The *Soil* mode is automatically selected when a Proctor value is enabled (see page 3–8).

**⚠ CAUTION**

When not taking measurements, always keep the source rod in the **SAFE** position. For added operator safety, the source rod on the gauge automatically retracts to the **SAFE** position when the gauge is lifted by the handle.

**⚠ Warning**

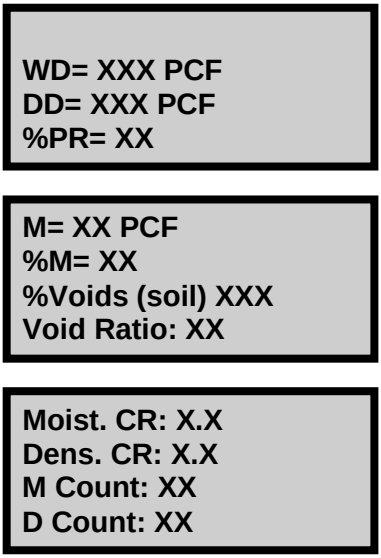
**Do not store or transport the gauge unless the sliding block is completely closed. Increased radiation levels may violate transportation regulations and cause excessive personnel exposure.**

Place the gauge on the test site. Release the gauge handle and push it down until it is in the correct position. Ensure that the handle stops in the notch designated for the proper measurement depth.

Press ⟨**START/ENTER**⟩**.**



After the count time has elapsed, the gauge displays the measurement results in a series of three screens, as follows. Use the up and down arrows to scroll through the various screens.




where:

***WD*** = Wet density in kg/m$^3$ or pcf

***DD*** = Dry density in kg/m$^3$ or pcf

***%PR*** = Percent Proctor (This value is valid only if an appropriate target has been entered for the material being tested.)

***MOIST*** = Moisture value in kg/m$^3$ or pcf

***% MOIST*** = Percent moisture

***Void Ratio*** = See description below

***Moist. CR*** = Moisture count ratio

***Dens. CR*** = Density count ratio

***M Count*** = Moisture counts as read by the gauge

***D Count*** = Density counts as read by the gauge

Press ⟨**START/ENTER**⟩ to return to the *Ready* mode.

Figure 11 illustrates the terms *void ratio* and *% air voids*. The *void ratio* is the ratio of the volume occupied by air and water in the soil to the volume occupied by solid particles. The term *% air voids* refers to the volume of air voids only as a percentage of the total volume.

The following formulas are used to calculate the *% air voids* and *void ratio* values.

$$\%\ AIR\ VOIDS = 100\ (1 - (Vs/Vt) - (Vw/Vt))$$

where:

***Vs*** = Volume of Soil

***Vt*** = Total Volume

***Vw*** = Volume of Water

or,

$$\%\ AIR\ VOIDS = 100\ (1 - (DD / SG(Dw)) - (M / (Dw)))$$

where:

***Dw*** = Density of Water

***SG*** = Specific Gravity of Soil Particles

***DD*** = Dry Density

***M*** = Moisture

***VOID RATIO* = Volume of Voids / Volume of Soil**

**= *( SG(Dw) – DD ) / DD***

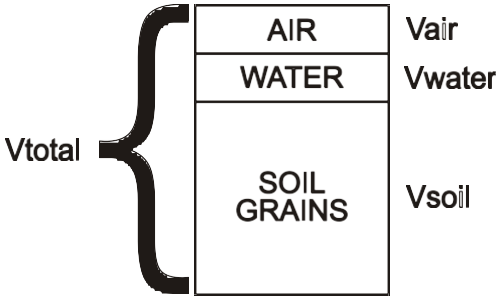



Figure 11. Voids Illustration

# Taking a Measurement – Asphalt Mode

The *Asphalt* mode is automatically selected when a Gmb value is enabled (see page 3–8).

**⚠ Warning**

**Do not store or transport the gauge unless the sliding block is completely closed. Increased radiation levels may violate transportation regulations and cause excessive personnel exposure.**

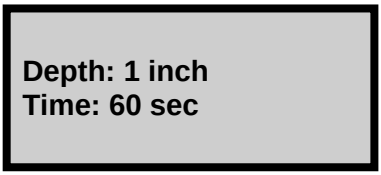
**⚠ CAUTION**

When not taking measurements, always keep the source rod in the **SAFE** position. For added operator safety, the source rod on the gauge automatically retracts to the **SAFE** position when the gauge is picked up by the handle.

**⚠ CAUTION**

The Model 3430 gauge is not waterproof. Please do not use it in the rain. If the gauge gets wet, make sure it is completely dry before sealing it in the case for storage. See page A-2 for an explanation.

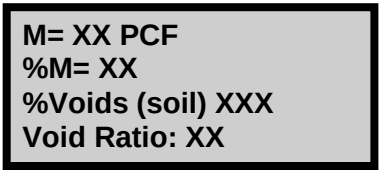
1. Place the gauge on the test site.
2. Release the gauge handle and push it into the backscatter position. Ensure that the handle is securely seated by gently tapping the handle down. Do not pass the notch.
3. Set the depth to *Backscatter* (1 inch) using the arrows on the keypad.
4. Press ⟨**START/ENTER**⟩.

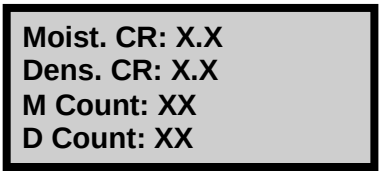



After the count time has elapsed, the gauge displays the measurement results in a series of three screens, as follows.

5. Use the up and down arrows to scroll through thevarious screens.



where:

***WD*** = Wet density in kg/m$^3$ or pcf

***DD*** = Dry density in kg/m$^3$ or pcf

***%Gmb =*** Bulk specific gravity

***MOIST*** = Moisture value in kg/m$^3$ or pcf

***% MOIST*** = Percent moisture

% VOIDS = 100 (1 – WD/VOIDLESS (Gmm))

***VOIDS RATIO*** = Value given by subtracting the percent Gmb value from 100. Applies to soil mode only.

***Moist. CR*** = Moisture count ratio

***Dens. CR*** = Density count ratio

***M Count*** = Moisture counts as read by the gauge

***D Count*** = Density counts as read by the gauge

6. Press ⟨**YES**⟩ to return to the *Ready* mode.